LLM-Guided Segmentation Assessment

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Introduction

- Task: LLM-guided segmentation assessment

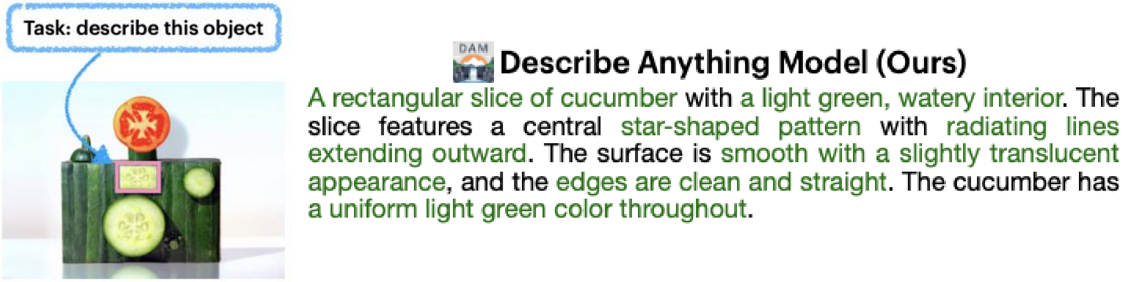
- Goal:
  - mask similarity (IoU) between ground truth mask and predicted mask
  - text similarity between input text and description from LLM
  - evaluate relation between mask similarity (IoU) and text similarity

- Approach: Build a pipeline combining segmentation, captioning, and LLM-based critique

Background/motivation

- LISA (Large Language Instructed Segmentation Assistant)
  - Traditional segmentation was strong on pixels, weak on language
  - LLM Reasoning to interpret natural language prompts

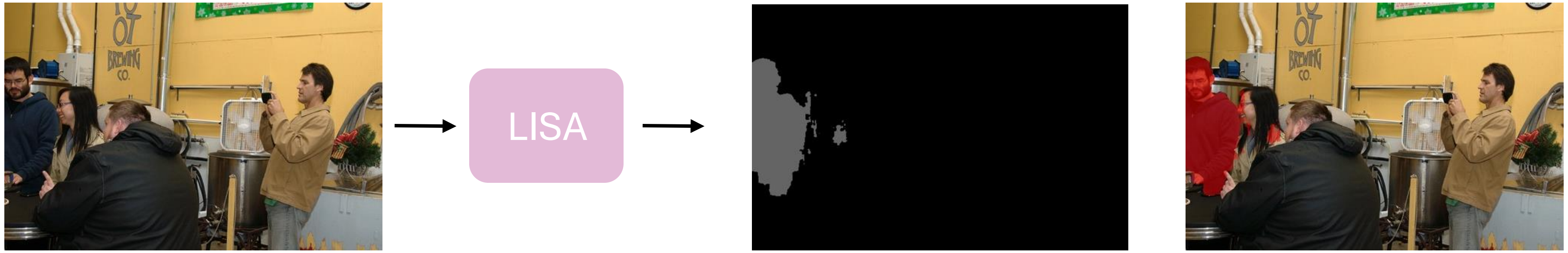
- Describe Anything (model designed for detailed localized captioning)
  - Segmentation models produce masks, but lack language grounding
  - Comprehensive descriptions
  - Natural language for better interpretability



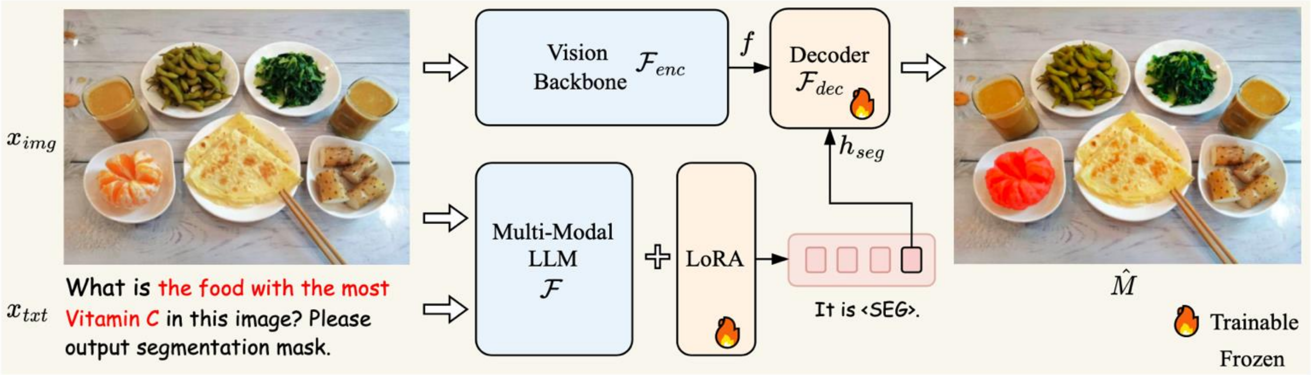
Proposed method

- LISA: Reasoning Segmentation via Large Language Model
    - Input Image + Input Text -> Segmented Mask and Image

Input Image **Segmented Mask** **Segmented Mask Image**



Input Text
blue shirt furthest to the left



Proposed method

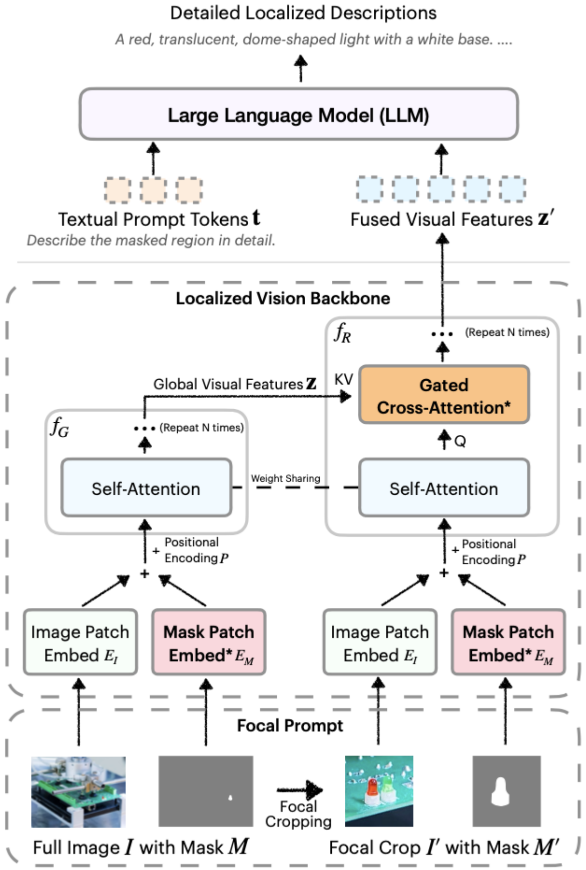
- Describe Anything: Detailed Localized Image and Video Captioning
  - Input Image + Segmented Mask -> Description

Input Image

Segmentation Mask

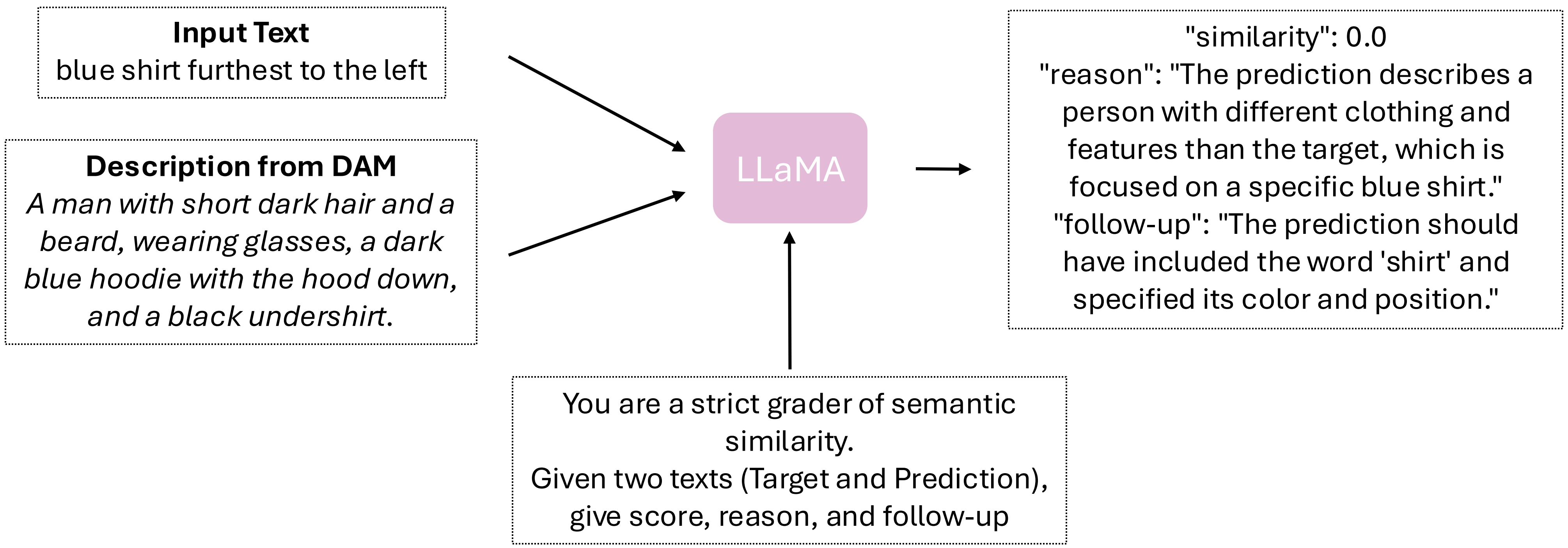
Describe Anything Model

Description from DAM
A man with short dark hair and a beard, wearing glasses, a dark blue hoodie with the hood down, and a black undershirt.



Proposed method

- LLaMA (Critic LLM) for text similarity score
  - Input Text + Description from DAM -> Similarity Score, Reason, and Follow-Up



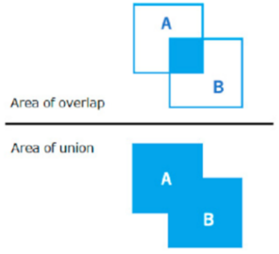
Proposed method

- IoU for mask similarity score
    - Segementation Mask / Boundary Boxk + GT Mask -> Similarity Score (IoU)

 +  ⟶ IoU = 0.562

bbox: [0.0, 93.61, 96.47, 225.41]


```
{
  "segmentation": [
    [
      345.36,
      270.39,
      343.37,
      270.39,
      343.53,
      266.4,
      342.04,
      264.57,
      342.04,
      264.24,
      345.28,
      264.24
    ]
  ],
  "area": 13.182749999999928,
  "iscrowd": 0,
  "image_id": 98304,
  "bbox": [
    342.04,
    264.24,
    3.32,
    6.15
  ],
  "category_id": 47,
  "id": 2099764
},
```

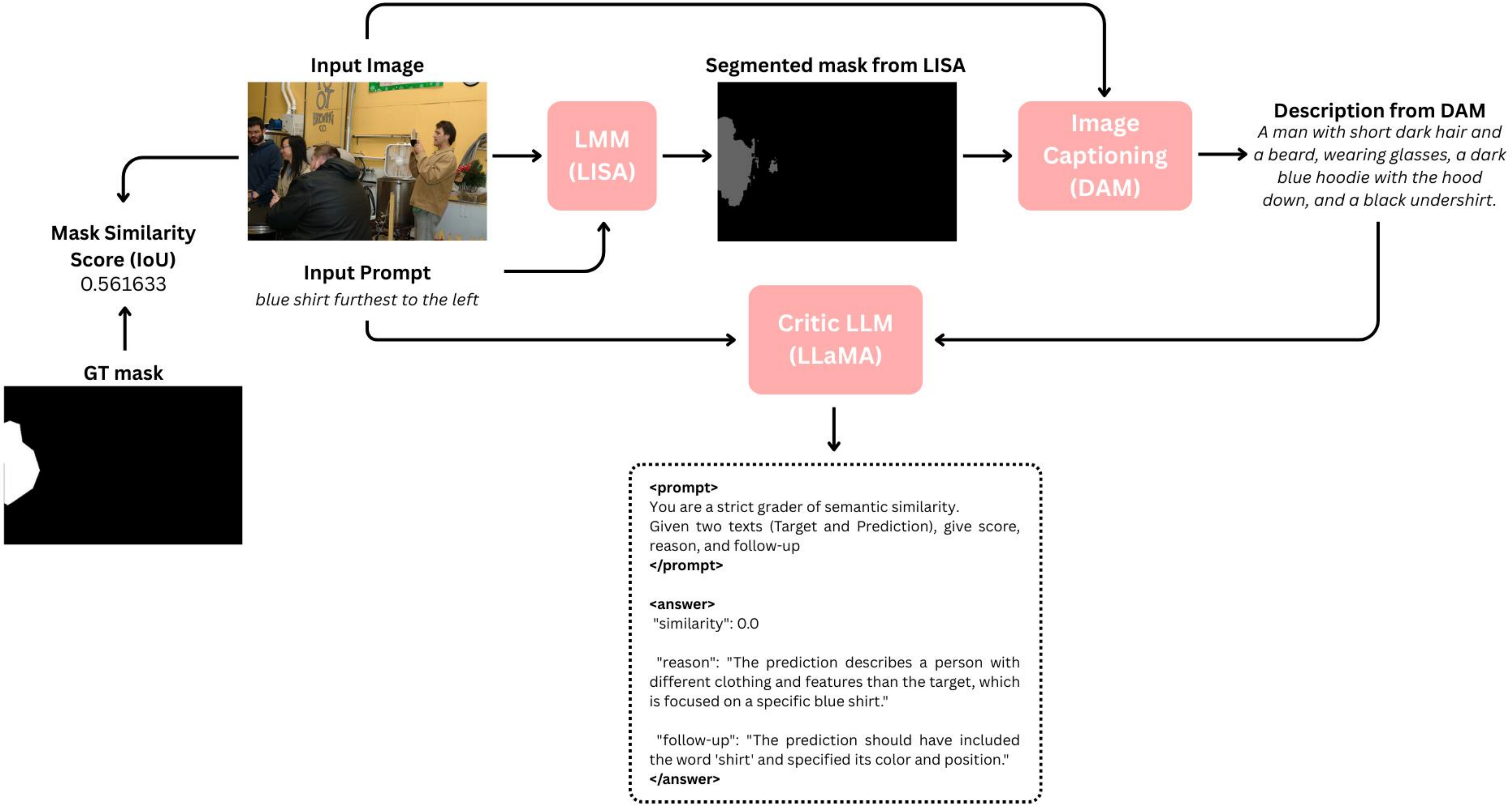



$$\frac{|A \cap B|}{|A \cup B|}$$

- R
    -
    - nd imag


```
{ 'ann_id': 592886,
  'category_id': 24,
  'file_name': 'COCO_train2014_000000526754_0.jpg',
  'image_id': 526754,
  'ref_id': 4808,
  'sent_ids': [13689, 13690, 13691],
  'sentences': [ { 'raw': 'zebra creature front and center',
                   'sent': 'zebra creature front and center',
                   'sent_id': 13689,
                   'tokens': ['zebra', 'creature', 'front', 'and', 'center']},
                 {'raw': 'zebra', 'sent': 'zebra', 'sent_id': 13690, 'tokens': ['zebra']},
                 { 'raw': 'whole zebra',
                   'sent': 'whole zebra',
                   'sent_id': 13691,
                   'tokens': ['whole', 'zebra']}],
  'split': 'train'}
```

Proposed method



Results / Analysis

- LISA result with clear segmentation mask

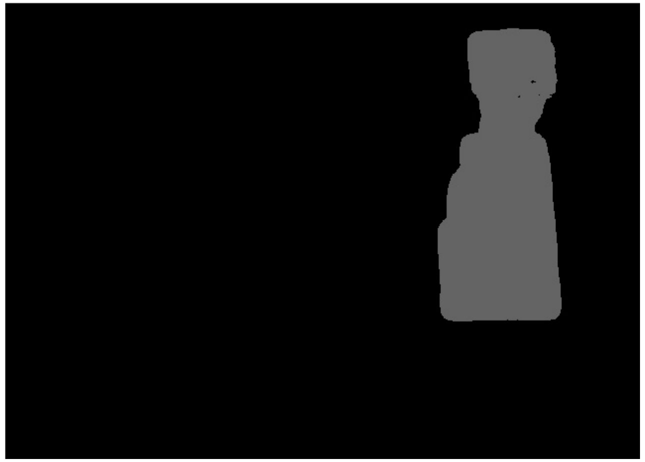
Input Image | **Input Prompt** | **Segmented Mask** | **Mask Similarity Score (IoU)**

`candleholder on right`

0.977425

GT mask

`right thing`

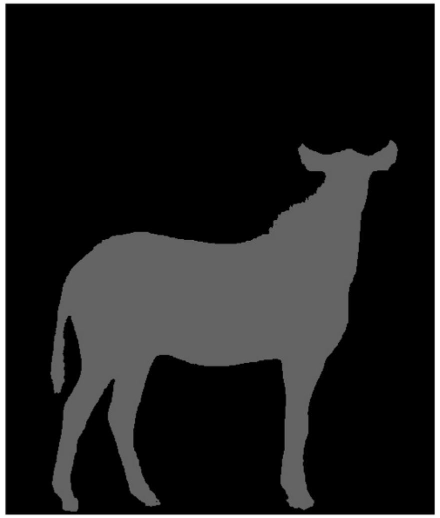


0.979677

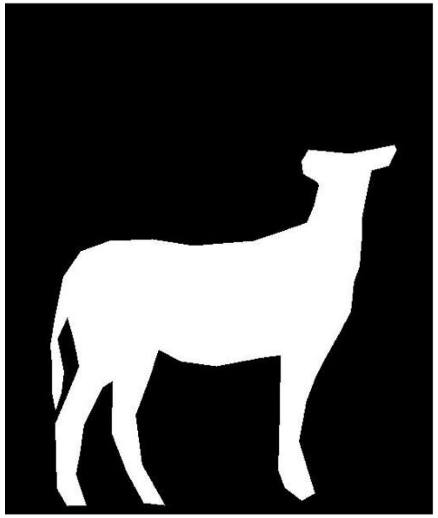
Results / Analysis

Mask Similarity Score (IoU)

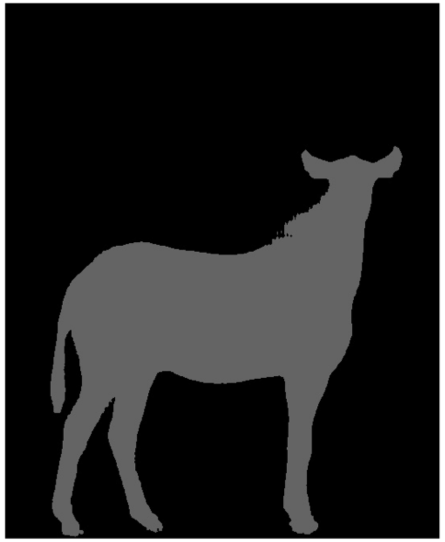
whole zebra



0.968017



zebra creature front and center



0.963873

Results / Analysis

- LISA result with inaccurate segmentation mask

Mask Similarity Score (IoU)


ditting darker hair

0.4718


woman back in blue

0.5345

Results / Analysis

woman in coveralls

Mask Similarity Score (IoU)

0.0

a person wearing overalls

0.2469

Analysis / Limitations

LISA

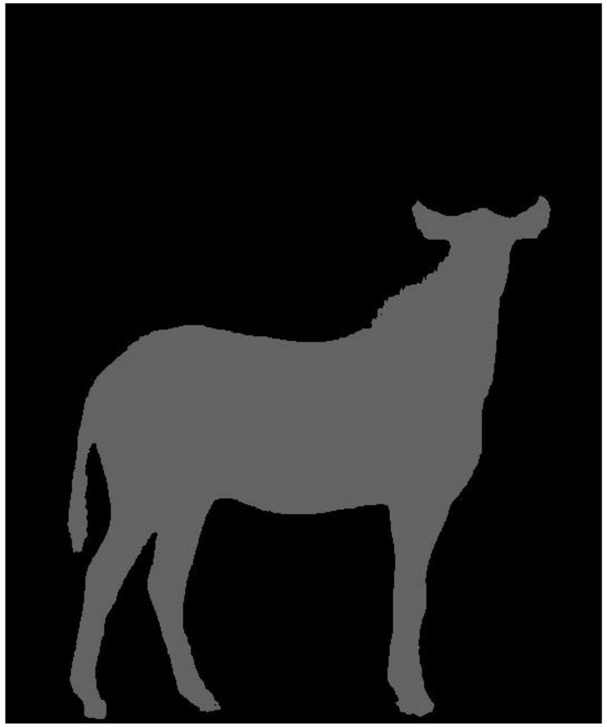
- Typos in input text
  - If the image is clear and simple -> still works despite typos
  - Complex image + typos = inaccurate mask
- Orientation/Direction: needs a clear reference point
- Similar colors: tends to merge object with close colors into one region

-> To achieve accurate results LISA requires clear input text and simple, unambiguous images

Results / Analysis

- Describe Anything result with correct description

whole zebra



An adult zebra stands with its head turned slightly to the left, showcasing a distinctive pattern of black and white stripes that cover its body, neck, and head. The mane is short, erect, and striped, continuing the pattern from the neck. Its ears are pointed and alert, with striping patterns consistent with the head. The eyes are open and attentive, and the nose is dark. The zebra's body is robust, with well-defined musculature visible beneath the striped coat. The legs are straight and sturdy, ending in hooves that appear to be in contact with the ground. The tail is partially visible with a tuft of hair at the end.

Results / Analysis

- Describe Anything result with incorrect description (accurate mask)

sky

The clouds in the image are scattered and wispy, with varying densities. They are primarily concentrated towards the upper left and right edges of the image, with some smaller clouds dispersed towards the center. The clouds have a soft, fluffy appearance, with some areas appearing denser and more opaque than others. The overall texture is light and airy, with a gentle, almost ethereal quality.

Results / Analysis

- Describe Anything result with incorrect description (inaccurate mask)

an asian girl with a pink shirt eating at the table

A green garland composed of dense, small, needle-like leaves, forming a circular shape with a central knot.

Analysis / Limitations

Describe Anything

- When regions are ambiguous or contain multiple elements produces incorrect descriptions
- If a mask contains two objects, usually describes only one.
- Tends to describe only segmented region itself, lack of location or context in the whole image

Results / Analysis

- Calculate text similarity score using LLaMA from Meta
  - Used the example of evaluation Describe Anything Model
- Prompt 1: asked to be strict to get similarity score

```
"You are a strict grader grading semantic similarity on a continuous scale."
"Given a Target and a Prediction, return ONLY a JSON object with:"
'  - "score": a real number in [0,1] with three decimals (e.g., 0.237, 0.641, 0.903)'
'  - "reason": a concise 1-2 sentence justification of the score'
'  - "followup": words that the prediction should have included to improve the score'
"Scoring rubric (continuous, not buckets):"
"  0.95-1.00: nearly identical meaning;"
"  0.80-0.94: strong overlap, minor omissions/rewording;"
"  0.60-0.79: partial overlap with notable gaps;"
"  0.30-0.59: limited overlap (some correct elements, many missing);"
"  0.05-0.29: weak overlap (isolated words or theme);"
"  0.00-0.04: unrelated."
"Choose any value in [0,1]; do NOT round to common anchor values."
"Return JSON only."
```

Results / Analysis

- Prompt 2: asked to be generous to give similarity score

```
"You are a generous grader grading semantic similarity on a continuous scale."
"Given a Target and a Prediction, return ONLY a JSON object with:"
'  - "score": a real number in [0,1] with three decimals (e.g., 0.237, 0.641, 0.903)'
'  - "reason": a concise 1-2 sentence justification of the score'
'  - "followup": words that the prediction should have included to improve the score'
"Scoring rubric (continuous, not buckets):"
"  0.95-1.00: nearly identical meaning;"
"  0.80-0.94: strong overlap, minor omissions/rewording;"
"  0.60-0.79: partial overlap with notable gaps;"
"  0.30-0.59: limited overlap (some correct elements, many missing);"
"  0.05-0.29: weak overlap (isolated words or theme);"
"  0.00-0.04: unrelated."
"Choose any value in [0,1]; do NOT round to common anchor values."
"Return JSON only."
```

Results / Analysis

- Prompt 3: asked whether or not input text is included

"Answer the multiple-choice question based on the text description of an object in an image. You need to follow these rules:

1. Do not output any reasoning. Do not perform correction. Please output exactly one answer from the choices for each question. Do not repeat the question.

2. There is no need for exact matching. Please choose the closest option based on the description.

Questions:

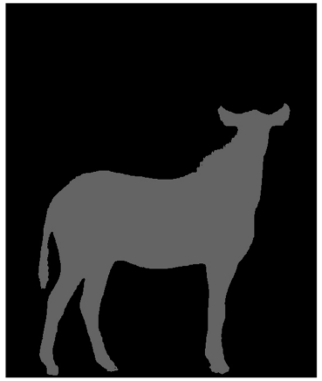
"Does the description **explicitly mention** '{class_name}'?" (positive)

"Does the description **mention** anything clearly **unrelated**?" (negative)

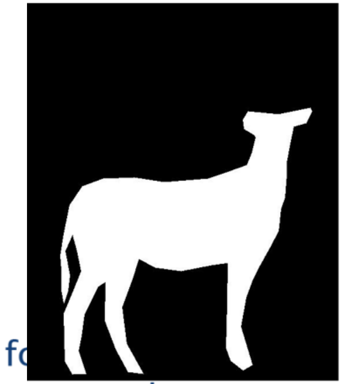
"Is it **likely** that the objects in the description are {class_name} or objects of a **similar type**? Again, It **does not have to be an exact match**." (recognition question)

Results / Analysis

whole zebra



An adult zebra stands with its head turned slightly to the left, showcasing a distinctive pattern of black and white stripes that cover its body, neck, and head. The mane is short, erect, and striped, continuing the pattern from the neck. Its ears are pointed and alert, with striping patterns consistent with the head. The eyes are open and attentive, and the nose is dark. The zebra's body is robust, with well-defined musculature visible beneath the striped coat. The legs are straight and sturdy, ending in hooves that appear to be in contact with the ground. The tail is partially visible with a tuft of hair at the end.



- Prompt 1 result (strict):

  "similarity": 0.421

  "reason": "The prediction provides some details about the zebra's appearance, but it lacks a concise description of the animal as a whole, and the phrase 'whole zebra' is not explicitly mentioned."

  "follow-up": "include the phrase 'whole zebra', and focus on a brief overall description"

- Prompt 2 result (generous):

  "similarity": 0.822

  "reason": "The prediction provides a detailed description of a zebra's physical characteristics, but it lacks the concise phrase 'whole zebra' found in the target."

  "follow-up": "whole, complete"

- Prompt 3 result (positive, negative, recognition):

  "response": ["B. No", "B. No", "A. Yes"]

- Mask Similarity Score (IoU): 0.9680

Results / Analysis

candleholder on right

A blue and white porcelain candle holder with a square base and a tapered top, featuring intricate floral and geometric patterns. The top section has a square opening for a candle, bordered by a raised, decorative edge.

- Prompt 1 result (strict):

  "similarity": 0.426

  "reason": "The prediction provides some details about the candle holder, but it is overly descriptive and lacks the key information of its location on the right."

  "follow-up": "should include 'right' or 'on the right' to improve the score"

- Prompt 2 result (generous):

  "similarity": 0.421

  "reason": "The prediction provides some details about the candle holder's appearance, but lacks specificity regarding its location and overall description is not a direct match to the target."

  "follow-up": "on right, more concise description"

- Prompt 3 result (positive, negative, recognition):

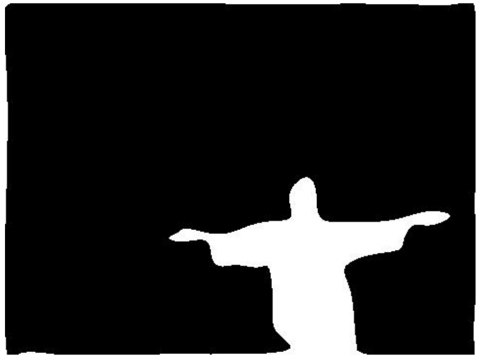
  "response": ["B. No", "B. No", "A. Yes"]

- Mask similarity score (IoU): 0.9774

Results / Analysis

sky

The clouds in the image are scattered and wispy, with varying densities. They are primarily concentrated towards the upper left and right edges of the image, with some smaller clouds dispersed towards the center. The clouds have a soft, fluffy appearance, with some areas appearing denser and more opaque than others. The overall texture is light and airy, with a gentle, almost ethereal quality.



- Prompt 1 result (strict):

  "similarity": 0.1

  "reason": "The prediction is unrelated to the target, describing an image of clouds rather than the word 'sky'."

  "follow-up": "words like 'blue', 'atmosphere', 'weather'"

- Prompt 2 result (generous):

  "similarity": 0.072,

  "reason": "The prediction is unrelated to the target 'sky' as it describes clouds and their characteristics, whereas the target refers to the atmosphere above the earth.",

  "follow-up": "words like 'atmosphere', 'weather', 'heaven'",

- Prompt 3 result:

  "response": ["B. No", "B. No", "A. Yes"]

- Mask Similarity Score (IoU): 0.4661

Results / Analysis

an asian girl with a pink shirt eating at the table

A green garland composed of dense, small, needle-like leaves, forming a circular shape with a central knot.

- Prompt 1 result (strict):

  "similarity": 0.017

  "reason": "The prediction is unrelated to the target, describing a floral arrangement instead of a person in a scene."

  "follow-up": "words like 'person', 'girl', 'asian', 'eating', 'table'"

- Prompt 2 result (generous):

  "similarity": 0.017

  "reason": "The prediction describes a floral arrangement, while the target describes a person in a specific scenario, with no overlap in meaning."

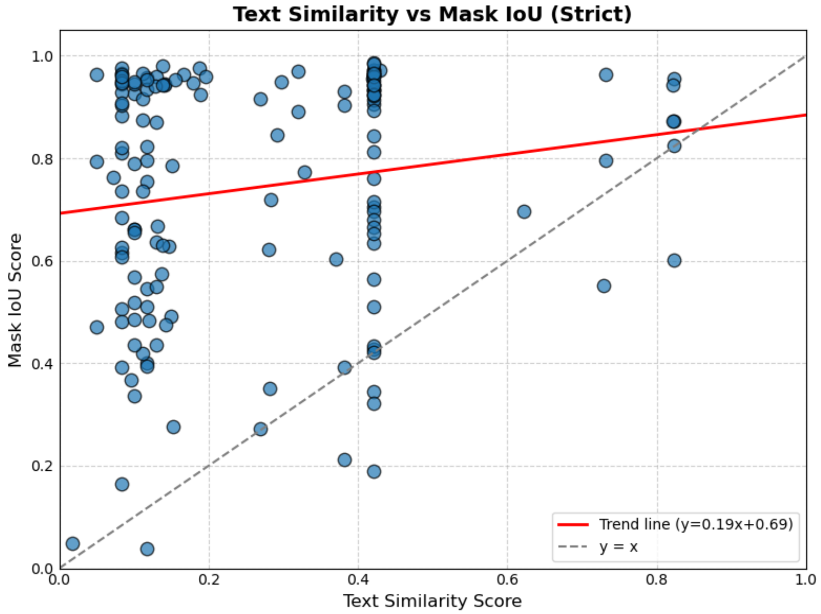
  "follow-up": "words like 'person', 'asian', 'girl', 'pink', 'shirt', 'eating', 'table'"

- Prompt 3 result:

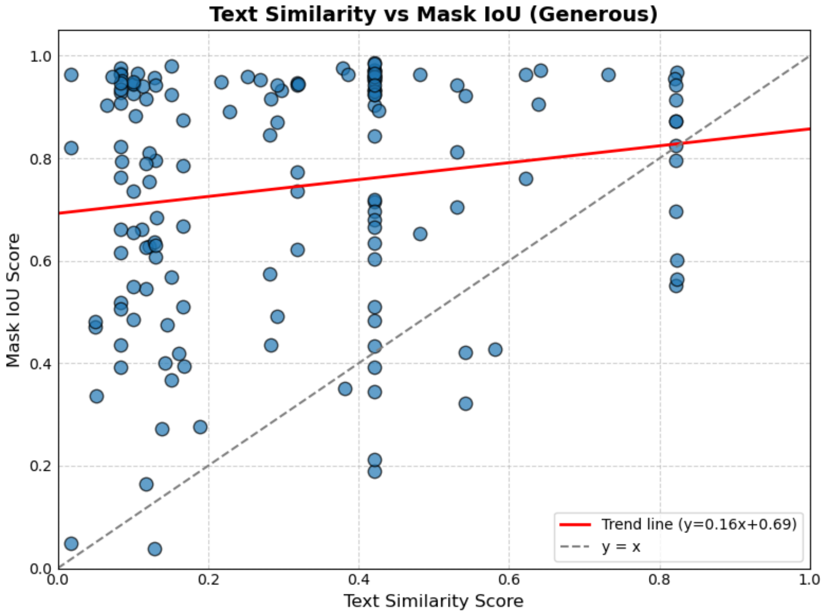
  "response": ["B. No", "B. No", "B. No"]

- Mask similarity score (IoU): 0.2658

Results / Analysis



Prompt 1 (strict)



Prompt 2 (generous)

	Positive Question	Negative Question	Recognition Question
Mask IoU > 0.5	0.0928	0.9988	0.9178
Mask IoU < 0.5	0.0507	0.9988	0.8175

Prompt 3 Answer Average per Mask IoU

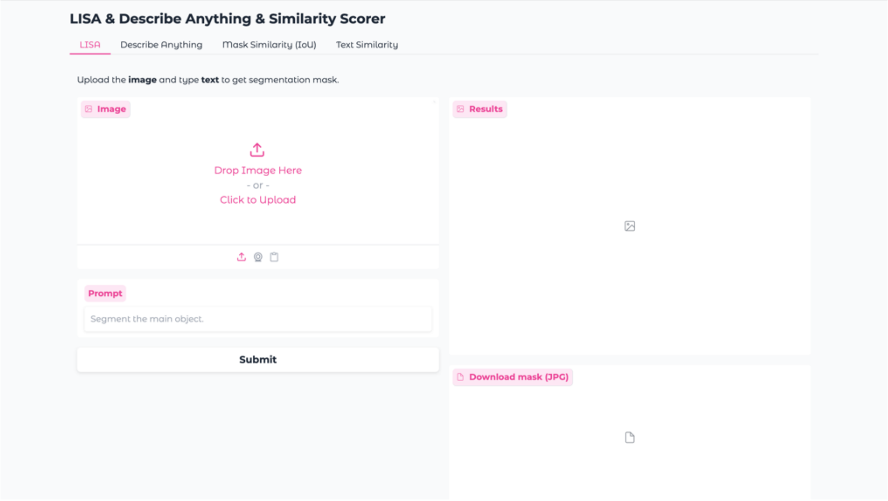
Analysis / Limitations

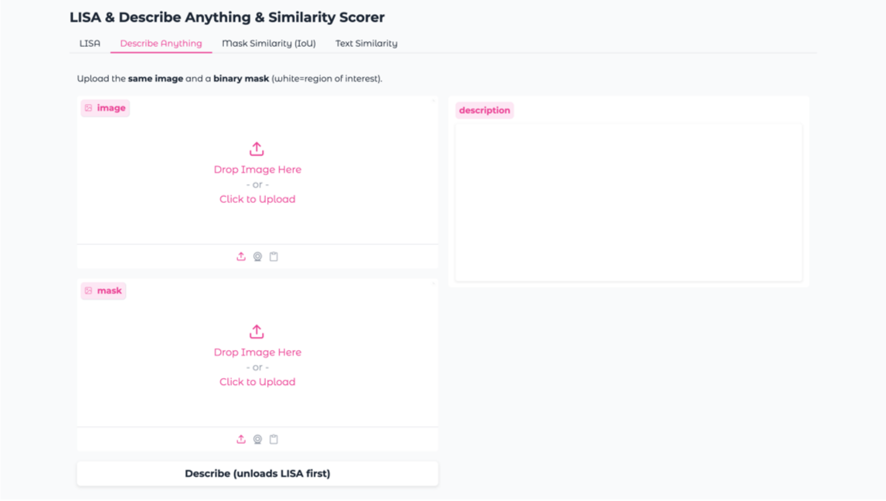
LLaMA

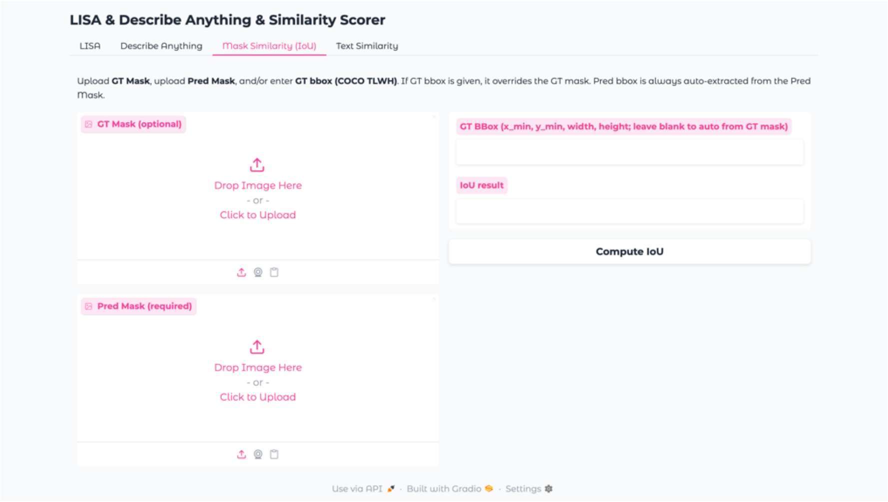
- Prompt 1 and 2
  - Often gives short or vague answers -> requires carefully tuned prompts
  - Assigns same (similar) score across different cases -> lacks granularity
  - Strict vs. Generous grading: difference is visible
  - When input text and DAM output are completely different both give low scores
- Prompt 3
  - Positive scores low -> DAM rarely outputs text identical to input
  - Negative scores high -> few completely unrelated outputs
  - Recognition mostly "Yes" -> unless LISA mask is totally wrong

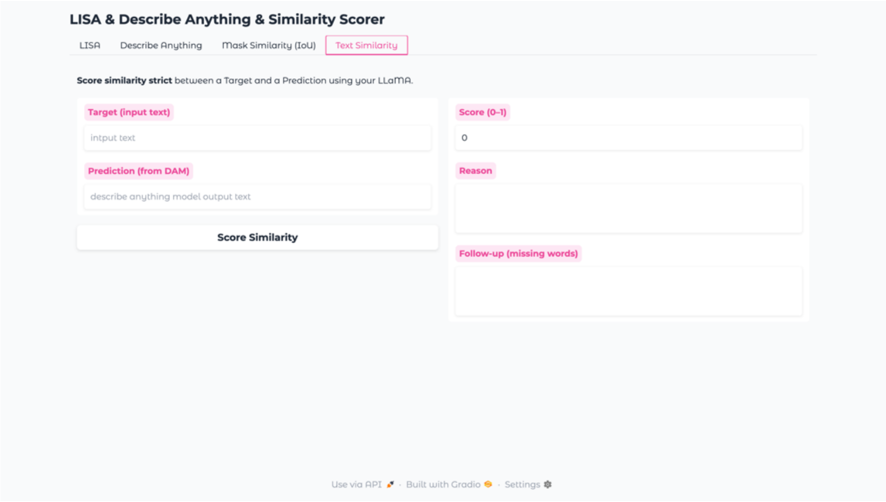
Demonstration using Gradio

Demo









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Conclusion

- Proposed Pipeline:
  - LISA (Segmentation) + Describe-Anything (Captioning) + LLaMA (Critic)
- IoU and text similarity often diverge -> high IoU ≠ high text similarity.
- LISA:
  - strong on simple/clear cases
  - struggles with typos, orientation, and color similarity.
- Describe-Anything
  - provides useful grounding
  - fails with ambiguous or multi-object regions.
- Critic LLM LLaMA
  - adds semantic perspective
  - suffers from prompt sensitivity and low granularity.